

Do the 5 Big Tech Companies, Apple, Amazon, Alphabet, Meta and Microsoft have THE largest data centers/server farms in the world? How do environmentalists calculate the carbon footprint of a data center?

Posted by u/DigitalFidgetal • 0 points • 17 comments

Are there algorithms to calculate the carbon footprint of a data center?

Examples:

A data center in a colder region will consume less power for cooling, lowering it's carbon footprint.

A data center that is geographically closer to it's end user base is likely to have a lower carbon footprint than one that is on a another continent.

Comments

u/gliffy • 2 points • 2023-05-21 18:35 UTC

Well Amazon alphabet and Microsoft are the biggest cloud providers so yes they have the biggest data centers. Amazon doesn't use coolers tho. Older sites my be chilled or use swamp coolers, newer sites use ambient air.

u/marmata75 • 2 points • 2023-05-22 06:06 UTC

Might be some of their own builds use ambient air, especially on colder climates. Definitely not the standard, even for new sites, especially if not built by them!

u/gliffy • 1 points • 2023-05-22 07:28 UTC

How many have you been in?

u/marmata75 • 1 points • 2023-05-22 11:51 UTC

As a worker in the Colo and hyper scale industry since many years, I know quite a few. In some never entered, but in a way or the other I've seen the designs, perhaps even be involved in the design. Is a small world! 🤔

u/gliffy • -1 points • 2023-05-22 12:29 UTC

Unless your in some niche area Amazon doesn't use colos.

u/scuppasteve • 2 points • 2023-05-22 13:59 UTC

Amazon's entire SFO and SEA clusters are in colos

u/gliffy • 0 points • 2023-05-23 03:03 UTC

Both are tiny and niche markets, when I went to SFO it was like 15 mw total, west 1 is pdx and exceeds 100mw

u/scuppasteve
• 3 points •
2023-05-23
04:33 UTC

Amazon has at least 200MW in colos in the US. Is it a huge percentage in total? No. But they are installed in a larger number of colos in the US than in their own

facilities. So let's not be silly. I used to run a good chunk of those Boardman facilities.

u/SlideFire • 2 points • 2023-05-22 13:13 UTC

Lol. You mean every single area outside of the continental US?

u/gliffy • 1 points • 2023-05-22 13:41 UTC

Ya basically, that's the 3rd world.

u/marmata75 • 2 points • 2023-05-22 13:01 UTC

Unless with 'niche area' you mean 'outside North America' then they do use third parties colo. The whole hyper scale industry is using them, Google, Microsoft, Oracle etc etc. There's an entire industry of datacenter operators that only builds for hyper scalers, specifically because they don't have the bandwidth to do builds everywhere they need to. Sometimes it's more effective just to lease!

u/DigitalFidgetal • 1 points • 2023-05-21 18:37 UTC

 lol

Data centers are sometimes compared to YEAST. Both are exothermic.

u/Redebo • 1 points • 2023-05-21 17:58 UTC

Your examples are not entirely representative. A data center in a colder climate doesn't necessarily use less power unless somehow the outside air is being used instead of mechanical cooling to handle the heat load.

Typically carbon emissions are calculated by three classes, Scope 1, 2, and 3. Scope one and two can be summarized as carbon released from the DC's own usage of equipment, energy, and their first party transportation emissions. Scope 3 are the emissions produced by their vendors in their service to the specific DC company.

There are software packages on the market that help DC operators classify, account for, and manage these three scopes. Schneider Electric is a company that offers this type of software.

Edit: Yes, the big five you specifically asked about do use software to account for their emissions. In addition several environmental watch dog groups do reporting on these companies and publish them regularly.

u/ghostalker4742 • 9 points • 2023-05-21 17:54 UTC

> Are there algorithms to calculate the carbon footprint of a data center?

I wouldn't call them algorithms, just formulas. It's based on physics after all, not user trends. At the most basic level: energy going in equals energy going out, as energy is never created or destroyed, just transformed. Then it's a matter of determining where you get the power from, what kind of fuels they use, and that help determines carbon footprint.

> A data center in a colder region will consume less power for cooling, lowering it's carbon footprint.

Look up free air return cooling. Here's [a good article](<https://journal.uptimeinstitute.com/data-center-free-air-cooling-trends/>) to get you started from Uptime Institute. It's several years old, but it's still a big, big topic in DC design and deployments. We're always chasing efficiencies after all.

> A data center that is geographically closer to it's end user base is likely to have a lower carbon footprint than one that is on a another continent.

I don't think there's a formula for that - it's premise alone seems false. Even if my customers are right down the road, I could host my environment in the cloud and they'd never notice. With high bandwidth, low latency connections, your data can live practically anywhere. The carbon footprint in a city 2000mi away could be less than the city next door, because they have different power providers, regulations, laws, tax incentives, etc.

u/DigitalFidgetal • 0 points • 2023-05-21 18:35 UTC

Do data centers need to pass some kinda EPA Environmental Protection Agency test? like cars, factories, etc. do?

u/Amalto • 1 points • 2023-05-31 19:49 UTC

It depends on the municipality but to the best of my understanding the closest thing is specifically for generator emissions. Different emissions standards are required based on how long the generators are supposed to run.